

Chuong Nguyen

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RESEARCH INTERESTS

Building AI agents that move from answering questions to expanding understanding, aligned with human interpretation. Interests: knowledge discovery and representation; curiosity-driven, abductive investigation; validating LLM interpretations against human judgment; agent systems design; resource-aware reinforcement learning.

EDUCATION

University of California, San Diego

B.S. in Computer Science, Department of Computer Science & Engineering | GPA: 3.803

Sep 2022 - Jun 2026

PUBLICATIONS

* denotes co-first authors.

Conference Papers

[2026] Xinyu Pi*, Qisen Yang*, **Chuong Nguyen***, Hua Shen. *Position: Bridge Human Interpretation and Machine Representation With Explicit Specification For Qualitative Data Analysis In LLM Era*. International Conference on Machine Learning (ICML), Position Paper Track, 2026. arXiv:2601.11739.

[2026] Taggart Smith*, **Chuong Nguyen***, Qisen Yang, Oishani Bandopadhyay, Yaru Su, Nadia Polikarpova, Xinyu Pi. *QualAlign: Benchmarking Automated Qualitative Coding Against Human Schemas*. Conference on Language Modeling (COLM), 2026. openreview.net/forum?id=FaPyCwiiTo.

[2024] Andrew Smithwick*, **Chuong Nguyen***, Emily Gorial*, Natasha Tran*, A. M. Flores, Imani N. S. Munyaka. *"Parent seeking Roblox Safety Help": Comparing Parental Roblox Concerns to Roblox Offerings*. IEEE International Symposium on Technology and Society (ISTAS), pp. 1-9, 2024.

Preprints

[2025] Xinyu Pi*, Qisen Yang*, **Chuong Nguyen***. *LOGOS: LLM-driven End-to-End Grounded Theory Development and Schema Induction for Qualitative Research*. arXiv preprint arXiv:2509.24294, 2025.

PRESENTATIONS

- Poster: QualAlign: Benchmarking Automated Qualitative Coding Against Human Schemas. COLM 2026, San Francisco, CA, Oct 2026.
- "Parent seeking Roblox Safety Help". Early Research Scholars Program Symposium, UC San Diego, 2024.
- Credit Card Fraud Detector. ACM AI Symposium, UC San Diego, 2023.

RESEARCH EXPERIENCE

Research Assistant, Prof. Zhiting Hu's Group - HDSI, UC San Diego

Apr 2025 - Present

Mentors: Xinyu (Frederick) Pi and Qiyue Gao (PhD students)

- LOGOS: co-developed and led implementation of an LLM-driven framework that turns unstructured evidence (interview transcripts, legal texts, agent trajectories) into hierarchical schemas, building semantic clustering, relation-aware graph reasoning, and iterative refinement; reaches an 88.2% matching rate against expert-developed schemas.
- QualAlign (COLM 2026): co-built a benchmark for automated qualitative coding and ran the baselines; organized evaluation metrics into lexical, semantic, and model-based interpretive categories, showing that metrics agree within a category but diverge across them.
- Position paper (ICML 2026): co-defined a framework crossing levels of meaning-making with levels of modeling, and mapped existing automated qualitative-analysis systems onto it.
- Episteme (in progress): core contributor to the agent harness for a deep research system, covering tool use, evidence synthesis, and evaluation.
- Fine-tuned LLMs with SFT and GRPO; explored test-time scaling and process reward models.

Research Assistant, Ujima Lab - UC San Diego

Sep 2023 - Oct 2024

Advisor: Prof. Imani Munyaka

- Beauty Filter: analyzed facial enhancements from three filter applications (Snatched, Blochi Gora, Baby Girl) to investigate cultural biases in appearance modification; migrated the facial landmark analysis pipeline to a remote server for parallel processing at scale.
- Smart Mirror: implemented DeepFace and Retina for facial recognition and racial classification on a Raspberry Pi

smart mirror, delivering real-time analysis for a public art installation on racial bias.

- Roblox Safety: co-first author on "Parent seeking Roblox Safety Help" (IEEE ISTAS 2024).

Early Research Scholars Program (ERSP) - UC San Diego

Sep 2023 - Jun 2024

- Conducted topic modeling on 10,000+ Reddit posts and surveyed 100 participants to identify four primary parental concerns on Roblox; proposed actionable platform-safety recommendations.
- Built a hate-crime news pipeline with SerpAPI, NewsPlease, DBSCAN, MeanShift, and the OpenAI API.

PROJECT EXPERIENCE

Software Developer, Zooseeker

Apr 2024 - Jun 2024

- Designed and implemented the database schema for a zoo navigation Android application.
- Revamped the Android UI and maintained project documentation for the development team.

ACM AI - UC San Diego

Oct 2022 - Jan 2023

Credit Card Fraud Detector

- Compared KNN, Random Forest, Decision Tree, SVM, and Logistic Regression for fraud detection.
- Built and presented a fraud detection web application at the ACM AI symposium.

TEACHING & MENTORING

Peer Mentor, UCSD Mentor Collective

2024

HONORS & AWARDS

- Provost Honors - UC San Diego
- Machine Learning Specialization - Coursera

LEADERSHIP & SERVICE

Volunteer Youth Leader, Vietnamese Eucharistic Youth Movement

2022 - Present

- Teach and mentor young children in the youth program.

SKILLS

Programming: Python, Java, C++, SQL

ML & Data: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, OpenCV, Streamlit

Research: NLP & LLMs, tool-using LLM agents, deep research agents, retrieval-augmented generation, knowledge representation, semantic clustering, graph-based reasoning, LLM evaluation, LLM fine-tuning (SFT, GRPO), reinforcement learning, data mining, topic modeling, qualitative methods

Tools: Git, Linux, LaTeX

Languages: English, Vietnamese